

Manoj Sunuguri

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PROFESSIONAL SUMMARY

Data Science and AI enthusiast with hands-on experience in Machine Learning, Deep Learning, NLP, Computer Vision, and Generative AI through industry internships and research projects. Skilled in Python, SQL, PyTorch, TensorFlow, and Scikit-learn, with experience building OCR, RAG, forecasting, and predictive analytics solutions for real-world applications.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

Tools & Frameworks: Machine Learning, Deep Learning, Natural Language Processing (NLP), Transformers, Probability & Statistics, Pandas, Scikit-learn, TensorFlow, PyTorch, Hugging Face, Numpy, Matplotlib, MySQL, LangChain, LangGraph.

Areas of Interest: Data Science, Computer Vision, Generative AI, Robotics, Problem Solving, Leadership

EXPERIENCE

Systems Intelligence Labs Pvt. Ltd.

May 2025 - July 2025

Deep Learning Engineer Intern

Hyd, India

- **Fine-tuned EasyOCR** (TPS-ResNet-BiLSTM-Attention) for **P&ID** drawings, achieving an **accuracy of 75%**.
- Built and deployed **YOLOv8-based object detection** and **segmentation** models, achieving **mAP@0.5 = 0.85**.
- Developed a **RAG** pipeline with **FAISS** and **LangChain**, enabling an **85% accurate** document query system.
- **Tech Stack:** PyTorch, OpenCV, YOLOv8, LangChain, FAISS, EasyOCR

Changwon National University, South Korea

Nov 2024 - Jan 2025

Research Internship

South Korea

- **Researched** and implemented Distinct Modular Pruning in **ResNet-Transformer** models.
- Achieving a **38% reduction** in model parameters while **preserving** over **95%** of baseline **image captioning accuracy**.
- Applied **pruning techniques** on the **UCM-Captions** dataset (21 classes, **2,100 images**) to optimise transformer-based image captioning for **geospatial analysis**. Achieving a **BLEU-4** score of **0.52**.
- **Tech Stack:** Model Pruning, Transformers, Image Captioning, Remote Sensing Data, METEOR, CIDEr

PROJECTS

Reddit Post Moderation Project

Feb 2025 - Mar 2025

- Applied **SQL-based EDA** with **PandaSQL** to uncover **patterns** in **Reddit post** ratings, deletions, and flags.
- Built an **NLP pipeline** with **TF-IDF** and preprocessing to train **Decision Tree & Random Forest models (F1: 0.72)**.
- Enhanced classification by integrating **Gemini 2.0 Flash** via **API**, achieving **91% accuracy** with **structured** outputs.
- **Tech Stack:** SQL, PandaSQL, TF-IDF, NLP Preprocessing, Decision Trees, Random Forest

Electrical Load Forecasting

Feb 2024 - Apr 2024

- Formulated **Web Scraping** scripts using **BeautifulSoup** to extract **load value data** from the **Indian Government** website.
- **Automated** data collection using **Cron Jobs** and stored the extracted data in **MongoDB**.
- **Processed** and optimised data for **Machine Learning** model training, developing **Linear Regression** and **Random Forest** models that achieved an **R² score of 0.96** for accurate **state load value predictions**.
- **Tech Stack:** BeautifulSoup, Cron Jobs, MongoDB, Data Preprocessing, Linear Regression, Random Forest

EDUCATION

Indian Institute of Technology (BHU) Varanasi, India

2026

Bachelor of Technology in Electrical Engineering

CGPA: 8.68/10.0

Narayana Junior College, Hyderabad

2022

TSBIE (XII)

Percentage: 97.9/100

Scholars International School, Hyderabad

2020

CBSE (X)

Percentage: 91.2/100

ACHIEVEMENTS

Dream11 | ML Scientist

Secured a **full time offer** on **Day 1 Slot 1** of the campus placements at IIT BHU, **offer rescinded** due to corporate restructuring process.

ML Forge Hackathon

Secured **1st Prize** among **100+ teams** in a machine learning challenge.

Flipr Hackathon 27.1

Won **Bronze Rank** for innovative generative AI solutions

Academic Excellence Award

Earned branch change to Electrical Engineering – **Top 1%**

Zelestra x AWS Hackathon

Finished in **Top 2% (20/1000+ Teams)** in the AWS ML Hackathon